# 数仓建模（ODS→DWD→DWS→ADS）

## 数据库

```
create database if not exists marketing_dw;
use marketing_dw;
```

## ODS 行为表

```
create external table if not exists ods_user_behavior (
    event_time      string comment '行为时间',
    event_type      string comment '行为类型(view/cart/purch
ase)',
    product_id      string comment '商品ID',
    category_id     string comment '品类ID',
    category_code   string comment '品类编码',
    brand           string comment '品牌',
    price           double comment '商品价格',
    user_id         string comment '用户ID',
    user_session    string comment '会话ID'
)
partitioned by (dt string comment '月份分区')
row format delimited
fields terminated by ','
stored as textfile
tblproperties ("skip.header.line.count"="1");
```

```
load data local inpath '/data/raw/kaggle/2019-Oct.csv'
overwrite into table ods_user_behavior
partition (dt='2019-10');
```

```
select * from ods_user_behavior limit 10;

select count(*) from ods_user_behavior;
42448764
```

```
hive> select * from ods_user_behavior limit 10;
OK
2019-10-01 00:00:00 UTC view    44600062        2103807459595387724             shiseido        35.79   541312140       72d76fde-8bb3-4e00-8c23-a032dfed738c    2019-10
2019-10-01 00:00:00 UTC view    3900821 2053013552326770905     appliances.environment.water_heater     aqua    33.2    554748717       9333dfbd-b87a-4708-9857-6336556b0fcc
9-10
2019-10-01 00:00:01 UTC view    17200506        2053013559792632471     furniture.living_room.sofa              543.1   519107250       566511c2-e2e3-422b-b695-cf8e6e792ca8
9-10
2019-10-01 00:00:01 UTC view    1307067 2053013558920217191     computers.notebook      lenovo  251.74  550050854       7c90fc70-0e80-4590-96f3-13c02c18c713    2019-10
2019-10-01 00:00:04 UTC view    1004237 2053013555631882655     electronics.smartphone  apple   1081.98 535871217       c6bd7419-2748-4c56-95b4-8cec9ff8b80d    2019-10
2019-10-01 00:00:05 UTC view    1480613 2053013561092866779     computers.desktop       pulser  908.62  512742880       0d0d91c2-c9c2-4e81-90a5-86594dec0db9    2019-10
2019-10-01 00:00:08 UTC view    17300353        2053013553853497655             creed   380.96  555447699       4fe811e9-91de-46da-90c3-bbd87ed3a65d    2019-10
2019-10-01 00:00:08 UTC view    31500053        2053013558031024687             luminarc        41.16   550978835       6280d577-25c8-4147-99a7-abc6048498d6    2019-10
2019-10-01 00:00:10 UTC view    28719074        2053013565480109009     apparel.shoes.keds      baden   102.71  520571932       ac1cd4e5-a3ce-4224-a2d7-ff660a105880    2019-10
2019-10-01 00:00:11 UTC view    1004545 2053013555631882655     electronics.smartphone  huawei  566.01  537918940       406c46ed-90a4-4787-a43b-59a410c1a5fb    2019-10
```

# DWD 全量清洗明细层

```
use marketing_dw;

drop table if exists dwd_user_behavior_detail;

create table dwd_user_behavior_detail (
    event_time_raw    string comment '原始行为时间',
    event_date        string comment '行为日期',
    event_hour        string comment '行为小时',
    event_type        string comment '行为类型',
    product_id        string comment '商品ID',
    category_id       string comment '品类ID',
    category_code     string comment '品类编码',
    brand             string comment '品牌',
    price             double comment '价格',
    user_id           string comment '用户ID',
    user_session      string comment '会话ID'
)
stored as orc;
```

```
insert overwrite table dwd_user_behavior_detail
select
    event_time as event_time_raw,
    substr(event_time, 1, 10) as event_date,
    substr(event_time, 12, 2) as event_hour,
```

```
    lower(event_type) as event_type,
    product_id,
    category_id,
    category_code,
    brand,
    cast(price as double) as price,
    user_id,
    user_session
from ods_user_behavior
where dt='2019-10'
  and user_id is not null
  and product_id is not null
  and event_type in ('view','cart','purchase');
```

```
select count(*) from dwd_user_behavior_detail;
42448764

select * from dwd_user_behavior_detail limit 10;
2019-10-01 00:00:00 UTC 2019-10-01      00      view    446
00062        2103807459595387724            shiseido
35.79   541312140       72d76fde-8bb3-4e00-8c23-a032dfed738
c
2019-10-01 00:00:00 UTC 2019-10-01      00      view    390
0821 2053013552326770905   appliances.environment.water_hea
ter      aqua    33.2    554748717       9333dfbd-b87a-4708
-9857-6336556b0fcc
2019-10-01 00:00:01 UTC 2019-10-01      00      view    172
00506        2053013559792632471    furniture.living_room.s
ofa              543.1   519107250       566511c2-e2e3-422b
-b695-cf8e6e792ca8
2019-10-01 00:00:01 UTC 2019-10-01      00      view    130
7067 2053013558920217191   computers.notebook       lenovo
251.74  550050854       7c90fc70-0e80-4590-96f3-13c02c18c71
3
2019-10-01 00:00:04 UTC 2019-10-01      00      view    100
4237 2053013555631882655   electronics.smartphone   apple
1081.98 535871217       c6bd7419-2748-4c56-95b4-8cec9ff8b80
d
```

```
2019-10-01 00:00:05 UTC 2019-10-01      00      view    148
0613 2053013561092866779   computers.desktop        pulser
908.62  512742880       0d0d91c2-c9c2-4e81-90a5-86594dec0db
9
2019-10-01 00:00:08 UTC 2019-10-01      00      view    173
00353        2053013553853497655            creed   380.96
555447699       4fe811e9-91de-46da-90c3-bbd87ed3a65d
2019-10-01 00:00:08 UTC 2019-10-01      00      view    315
00053        2053013558031024687            luminarc
41.16   550978835       6280d577-25c8-4147-99a7-abc6048498d
6
2019-10-01 00:00:10 UTC 2019-10-01      00      view    287
19074        2053013565480109009    apparel.shoes.keds
baden   102.71  520571932       ac1cd4e5-a3ce-4224-a2d7-ff6
60a105880
2019-10-01 00:00:11 UTC 2019-10-01      00      view    100
4545 2053013555631882655   electronics.smartphone   huawei
566.01  537918940       406c46ed-90a4-4787-a43b-59a410c1a5f
b
Time taken: 0.109 seconds, Fetched: 10 row(s)
```

# DWS

## 用户行为汇总表dws_user_behavior_summary

```
drop table if exists dws_user_behavior_summary;

create table dws_user_behavior_summary (
    user_id                      string comment '用户ID',
    first_event_time             string comment '首次行为时
间',
    last_event_time              string comment '最近行为时
间',
    total_view_cnt               bigint comment '浏览次数',
    total_cart_cnt               bigint comment '加购次数',
    total_purchase_cnt           bigint comment '购买次数',
    total_purchase_amount        double comment '总消费金
```

```
额',
    active_day_cnt               bigint comment '活跃天数',
    avg_order_amount             double comment '平均订单金
额'
)
stored as orc;
```

```
insert overwrite table dws_user_behavior_summary
select
    user_id,
    min(event_time_raw) as first_event_time,
    max(event_time_raw) as last_event_time,
    sum(case when event_type='view' then 1 else 0 end) as t
otal_view_cnt,
    sum(case when event_type='cart' then 1 else 0 end) as t
otal_cart_cnt,
    sum(case when event_type='purchase' then 1 else 0 end)
as total_purchase_cnt,
    sum(case when event_type='purchase' then price else 0 e
nd) as total_purchase_amount,
    count(distinct event_date) as active_day_cnt,
    case
        when sum(case when event_type='purchase' then 1 els
e 0 end)=0 then 0
        else sum(case when event_type='purchase' then price
else 0 end)
             / sum(case when event_type='purchase' then 1 e
lse 0 end)
    end as avg_order_amount
from dwd_user_behavior_detail
group by user_id;
```

```
select count(*) from dws_user_behavior_summary;
3022290

select * from dwd_user_behavior_detail limit 10;
2019-10-01 00:00:00 UTC 2019-10-01      00      view    446
```

```
00062        2103807459595387724            shiseido
35.79   541312140       72d76fde-8bb3-4e00-8c23-a032dfed738
c
2019-10-01 00:00:00 UTC 2019-10-01      00      view    390
0821 2053013552326770905   appliances.environment.water_hea
ter      aqua    33.2    554748717       9333dfbd-b87a-4708
-9857-6336556b0fcc
2019-10-01 00:00:01 UTC 2019-10-01      00      view    172
00506        2053013559792632471    furniture.living_room.s
ofa              543.1   519107250       566511c2-e2e3-422b
-b695-cf8e6e792ca8
2019-10-01 00:00:01 UTC 2019-10-01      00      view    130
7067 2053013558920217191   computers.notebook       lenovo
251.74  550050854       7c90fc70-0e80-4590-96f3-13c02c18c71
3
2019-10-01 00:00:04 UTC 2019-10-01      00      view    100
4237 2053013555631882655   electronics.smartphone   apple
1081.98 535871217       c6bd7419-2748-4c56-95b4-8cec9ff8b80
d
2019-10-01 00:00:05 UTC 2019-10-01      00      view    148
0613 2053013561092866779   computers.desktop        pulser
908.62  512742880       0d0d91c2-c9c2-4e81-90a5-86594dec0db
9
2019-10-01 00:00:08 UTC 2019-10-01      00      view    173
00353        2053013553853497655            creed   380.96
555447699       4fe811e9-91de-46da-90c3-bbd87ed3a65d
2019-10-01 00:00:08 UTC 2019-10-01      00      view    315
00053        2053013558031024687            luminarc
41.16   550978835       6280d577-25c8-4147-99a7-abc6048498d
6
2019-10-01 00:00:10 UTC 2019-10-01      00      view    287
19074        2053013565480109009    apparel.shoes.keds
baden   102.71  520571932       ac1cd4e5-a3ce-4224-a2d7-ff6
60a105880
2019-10-01 00:00:11 UTC 2019-10-01      00      view    100
4545 2053013555631882655   electronics.smartphone   huawei
566.01  537918940       406c46ed-90a4-4787-a43b-59a410c1a5f
```

```
b
Time taken: 0.089 seconds, Fetched: 10 row(s)
```

# RFM 基础表

```
drop table if exists dws_user_rfm_base;

create table dws_user_rfm_base (
    user_id            string comment '用户ID',
    recency_days       int comment '最近一次购买距统计日天数',
    frequency_cnt      bigint comment '购买次数',
    monetary_amount    double comment '累计消费金额',
    last_purchase_time string comment '最近购买时间'
)
stored as orc;

insert overwrite table dws_user_rfm_base
select
    user_id,
    datediff('2019-10-31', max(event_date)) as recency_day
s,
    count(1) as frequency_cnt,
    sum(price) as monetary_amount,
    max(event_time_raw) as last_purchase_time
from dwd_user_behavior_detail
where event_type='purchase'
group by user_id;
```

## 查询

```
select count(*) from dws_user_rfm_base;
347118

select * from dws_user_rfm_base limit 10;
264649825       25      2       1240.04 2019-10-06 20:36:11
```

```
UTC
384989212       13      1       41.44   2019-10-18 11:19:59
UTC
396222093       7       1       48.14   2019-10-24 17:54:48
UTC
410333972       0       1       591.25  2019-10-31 12:37:15
UTC
419697235       9       1       57.66   2019-10-22 05:07:32
UTC
420935067       24      1       385.84  2019-10-07 08:39:16
UTC
427151532       15      1       177.35  2019-10-16 10:35:19
UTC
427391662       16      2       430.46  2019-10-15 18:32:30
UTC
434170823       5       3       2112.54 2019-10-26 08:51:02
UTC
434350078       2       1       128.45  2019-10-29 11:31:23
UTC
Time taken: 0.087 seconds, Fetched: 10 row(s)
```

# ADS：用户分群

```
drop table if exists ads_user_rfm_segment;

create table ads_user_rfm_segment (
    user_id             string comment '用户ID',
    recency_days        int comment '最近购买天数',
    frequency_cnt       bigint comment '购买频次',
    monetary_amount     double comment '消费金额',
    r_score             int comment 'R得分',
    f_score             int comment 'F得分',
    m_score             int comment 'M得分',
    user_segment        string comment '用户分群'
)
stored as orc;
```

```
insert overwrite table ads_user_rfm_segment
select
    user_id,
    recency_days,
    frequency_cnt,
    monetary_amount,
    case
        when recency_days <= 3 then 5
        when recency_days <= 7 then 4
        when recency_days <= 15 then 3
        when recency_days <= 30 then 2
        else 1
    end as r_score,
    case
        when frequency_cnt >= 20 then 5
        when frequency_cnt >= 10 then 4
        when frequency_cnt >= 5 then 3
        when frequency_cnt >= 2 then 2
        else 1
    end as f_score,
    case
        when monetary_amount >= 3000 then 5
        when monetary_amount >= 1500 then 4
        when monetary_amount >= 800 then 3
        when monetary_amount >= 300 then 2
        else 1
    end as m_score,
    case
        when recency_days <= 7 and frequency_cnt >= 5 and m
onetary_amount >= 1500
            then '高价值用户'
        when recency_days <= 15 and frequency_cnt >= 2 and
monetary_amount >= 300
            then '潜力用户'
        when recency_days > 20 and frequency_cnt <= 2
            then '流失风险用户'
        else '一般用户'
```

```
    end as user_segment
from dws_user_rfm_base;
```

## 查询

```
select user_segment, count(*) as user_cnt
from ads_user_rfm_segment
group by user_segment;

一般用户         202217
流失风险用户     78535
潜力用户         56792
高价值用户       9574
```

# ADS：可视化统计表

```
drop table if exists ads_user_segment_distribution;

create table ads_user_segment_distribution (
    user_segment      string comment '用户分群',
    user_cnt          bigint comment '用户数',
    total_amount      double comment '总消费金额',
    avg_amount        double comment '人均消费金额'
)
stored as orc;

insert overwrite table ads_user_segment_distribution
select
    user_segment,
    count(*) as user_cnt,
    sum(monetary_amount) as total_amount,
    avg(monetary_amount) as avg_amount
from ads_user_rfm_segment
group by user_segment;
```

## 查询

```
select * from ads_user_segment_distribution;
一般用户        202217  7.365241809000005E7     364.22466009
287075
流失风险用户    78535   2.4866215520000644E7    316.625905901
83544
潜力用户        56792   7.247880955999954E7     1276.2151281
870606
高价值用户      9574    5.896005909999997E7     6158.3516920
827215
```

# 用户分群表

```
use marketing_dw;

drop table if exists ads_user_rfm_score;

create table ads_user_rfm_score (
    user_id                  string comment '用户ID',
    total_view_cnt           bigint comment '总浏览次数',
    total_cart_cnt           bigint comment '总加购次数',
    total_purchase_cnt       bigint comment '总购买次数',
    total_purchase_amount    double comment '总消费金额',
    avg_order_amount         double comment '平均订单金额',
    active_day_cnt           bigint comment '活跃天数',
    user_value_level         string comment '用户价值分层',
    rfm_tag                  string comment 'RFM标签说明'
)
stored as orc;
```

## 查询

```
select * from ads_user_rfm_score limit 20;
195082191       1       0       0       0.0     0.0     1
流失风险用户    低活跃低转化
```

```
200673532       4       0       0       0.0     0.0     3
流失风险用户     低活跃低转化
216064734       5       0       0       0.0     0.0     2
流失风险用户     低活跃低转化
231943435       1       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
245978545       4       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
249183593       1       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
251924925       10      0       0       0.0     0.0     1
潜力用户          高浏览低转化
257849716       1       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
261006747       3       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
263428680       1       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
270033537       18      0       0       0.0     0.0     1
潜力用户          高浏览低转化
270844802       2       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
271831344       1       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
277067319       7       0       0       0.0     0.0     2
流失风险用户     低活跃低转化
280593902       1       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
282977436       24      0       0       0.0     0.0     3
潜力用户          高浏览低转化
286192318       1       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
291107595       1       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
292071852       2       0       0       0.0     0.0     2
流失风险用户     低活跃低转化
294641715       1       0       0       0.0     0.0     1
流失风险用户     低活跃低转化
```

```
Time taken: 0.087 seconds, Fetched: 20 row(s)

select user_value_level, count(*) as user_cnt
from ads_user_rfm_score
group by user_value_level;
一般用户        281204
流失风险用户    1999866
潜力用户        708653
高价值用户      32567
```

# 精准营销策略表

```
drop table if exists ads_precision_marketing_crowd;

create table ads_precision_marketing_crowd (
    user_id                  string comment '用户ID',
    user_value_level         string comment '用户分层',
    total_view_cnt           bigint comment '浏览次数',
    total_purchase_cnt       bigint comment '购买次数',
    total_purchase_amount    double comment '消费金额',
    marketing_strategy       string comment '营销策略'
)
stored as orc;

insert overwrite table ads_precision_marketing_crowd
select
    user_id,
    user_value_level,
    total_view_cnt,
    total_purchase_cnt,
    total_purchase_amount,
    case
        when user_value_level = '高价值用户' then '推送高端商
品、会员专属优惠和个性化推荐'
        when user_value_level = '潜力用户' then '发放优惠券、限
时折扣，促进首次转化'
        when user_value_level = '一般用户' then '推送常规促销和
```

```
热门商品推荐'
        when user_value_level = '流失风险用户' then '发送召回活
动信息，激活用户回访'
        else '推送常规营销信息'
    end as marketing_strategy
from ads_user_rfm_score;
```

## 查询

```
select * from ads_precision_marketing_crowd limit 10;
195082191       流失风险用户    1       0       0.0     发送召
回活动信息，激活用户回访
200673532       流失风险用户    4       0       0.0     发送召
回活动信息，激活用户回访
216064734       流失风险用户    5       0       0.0     发送召
回活动信息，激活用户回访
231943435       流失风险用户    1       0       0.0     发送召
回活动信息，激活用户回访
245978545       流失风险用户    4       0       0.0     发送召
回活动信息，激活用户回访
249183593       流失风险用户    1       0       0.0     发送召
回活动信息，激活用户回访
251924925       潜力用户        10      0       0.0     发放
优惠券、限时折扣，促进首次转化
257849716       流失风险用户    1       0       0.0     发送召
回活动信息，激活用户回访
261006747       流失风险用户    3       0       0.0     发送召
回活动信息，激活用户回访
263428680       流失风险用户    1       0       0.0     发送召
回活动信息，激活用户回访

select user_value_level, marketing_strategy, count(*) as us
er_cnt
from ads_precision_marketing_crowd
group by user_value_level, marketing_strategy;
一般用户        推送常规促销和热门商品推荐      281204
流失风险用户    发送召回活动信息，激活用户回访  1999866
```

| | | |
|---|---|---|
| 潜力用户 | 发放优惠券、限时折扣，促进首次转化 | 708653 |
| 高价值用户 | 推送高端商品、会员专属优惠和个性化推荐 | 32567 |